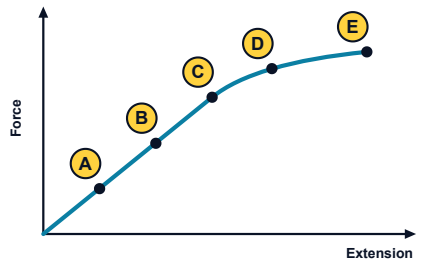
DO NOW · AO1 REVISION

Forces and elasticity

Forces
Paper 2 · Combined and Separate
20 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 9 starred (★) questions first.

	QUESTION	ANSWER	✓
1	**Give three ways in which the shape of an object can be changed.**	Stretching, compressing or bending	☐
2 ★	**What is elastic deformation?**	The object returns to its original shape when the forces are removed.	☐
3 ★	**What is inelastic deformation?**	The object does not return to its original shape when the forces are removed.	☐
4 ★	**State how the extension of a spring depends on the force applied, below the limit of proportionality.**	The extension is directly proportional to the force.	☐
5	**Describe the relationship between the force on an elastic object and its extension.**	As the force increases, the extension increases.	☐
6 ★	**What is the limit of proportionality of an elastic object?**	The point beyond which extension is no longer directly proportional to force.	☐
7	**Which letter shows the limit of proportionality on this force–extension graph?** 	C	☐
8	**State the standard unit of force.**	Newton (N)	☐
9 ★	**State the standard unit of spring constant.**	Newton per metre (N/m)	☐
10 ★	**Define spring constant.**	The force needed to stretch (or compress) a spring by 1 m.	☐

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	QUESTION	ANSWER	✓
11	**State the standard unit of extension.**	Metre (m)	☐
12 ★	**State how the extension of an elastic object is calculated.**	Extension = stretched length − original length	☐
13 ★	**Name the energy store that increases when an elastic object is stretched or compressed.**	The elastic potential energy store	☐
14 ★	**Describe how to find the spring constant from a graph of force (y-axis) against extension (x-axis).**	It is the gradient of the straight section.	☐
15	**Describe how to find the spring constant from a graph of extension (y-axis) against force (x-axis).**	It is 1 ÷ gradient of the straight section.	☐
16	**State the standard unit of elastic potential energy.**	Joule (J)	☐
17	**State the equation linking force, spring constant and extension.**	force = spring constant × extension ($F = k\,e$)	☐
18	**Explain why more than one force is needed to stretch or compress a stationary object.**	A single force would just make the object move.	☐
19	**State how the work done stretching a spring compares with the elastic potential energy stored.**	They are equal (if the spring is not inelastically deformed).	☐
20	**State how linear and non-linear relationships look on a force–extension graph.**	Linear: a straight line. Non-linear: a curve.	☐

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